LIQPLAT Statistical Analysis Plan

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1 Administrative information

1.1 Revision history

Version	Date	Who	Comments
0.1	2024-09-20	Johannes Schwenke	First draft
0.9	2025-10-27	Johannes Schwenke	Consolidated draft based on results from simulations studies

1.2 Roles and responsibilities

Name	Affiliation	Role
Giusi Moffa	University of Basel	Trial statistician
Johannes Schwenke	University of Basel and University Hospital of Basel	PhD Student
Benjamin Kasenda	University Hospital of Basel	Principal investigator and Sponsor-Investigator
Matthias Briel	University of Basel and University Hospital of Basel	TBD

1.3 Abbreviations

Abbreviation	Full Term
ATE	Average Treatment Effect
BSC	Best Supportive Care
CDWH	Clinical Data Warehouse
CHIP	Clonal Hematopoiesis of Indeterminate Potential
ctDNA	Circulating Tumor DNA
DAG	Directed Acyclic Graph
ECOG	Eastern Cooperative Oncology Group
EHR	Electronic Health Records
EORTC	European Organisation for Research and Treatment of Cancer
GRC	General Research Consent
HR	Hazard Ratio
IE	Intercurrent Event
IQR	Interquartile Range
ITT	Intention-to-Treat
KM	Kaplan-Meier
LOO-CV	Leave-One-Out Cross-Validation
MAR	Missing At Random
MCAR	Missing Completely at Random

Abbreviation	Full Term
MCMC	Markov Chain Monte Carlo
MICE	Multiple Imputation by Chained Equations
NA	Not Applicable / Nelson-Aalen
OR	Odds Ratio
OS	Overall Survival
PFS	Progression-Free Survival
PMM	Predictive Mean Matching
PO	Proportional Odds
PROMs	Patient Reported Outcome Measures
psATE	Principal Stratum Average Treatment Effect
QLQ-C15	Quality of Life Questionnaire - Core 15
QLQ-C30	Quality of Life Questionnaire - Core 30
QoL	Quality of Life
rcs	Restricted Cubic Splines
RCT	Randomized Controlled Trial
RMST	Restricted Mean Survival Time
SAP	Statistical Analysis Plan
SAT	Single-Arm Trial
SE	Standard Error
SoC	Standard of Care
SOP	State Occupancy Probability
TAOOH	Time Out of Hospital

2 Introduction

The purpose of this Statistical Analysis Plan (SAP) is to provide a detailed description of the intended statistical analyses for the LIQPLAT trial. The SAP is based on the protocol version 1.1.

We developed this SAP using data sets based on routinely collected data of patients who sought care prior to LIQPLAT, and who would have fulfilled the eligibility criteria. These data are complemented with simulations where needed.

3 Trial Overview

3.1 Background and rationale

Liquid biopsies, particularly circulating tumor DNA (ctDNA) analysis, offer potential solutions to challenges in treating patients with solid cancers. ctDNA may provide a more comprehensive view of tumor heterogeneity than traditional biopsies, aiding in identifying targetable alterations, quantifying disease burden, detecting early treatment resistance, and predicting outcomes. While promising, more robust trial data on the implementation of ctDNA in routine care is needed. We therefore conduct a single-arm trial (SAT) to investigate ctDNA implementation in routine care for advanced cancer patients, using random invitation to ensure representative sampling. The random invitation to participation allows for a randomized comparison with patients not invited to the trial.

3.2 Objectives

As per protocol LIQPLAT v. 1.1 has two primary objectives:

1. To assess the implementation and feasibility of ctDNA measurements from peripheral blood during routine clinical care of cancer patients in the University Hospital Basel.
2. To compare clinical and patient-reported outcomes of patients from this trial with patients who did not receive systematic ctDNA measurement.

This SAP focuses primarily on the second objective, detailing the analyses for the three primary outcomes of the randomized comparison: Overall Survival (OS), Quality of Life (QoL), and Time Out of Hospital (TAOOH).

4 Study methods

4.1 Trial design

This is a single-arm, single-center trial at the University Hospital of Basel. Patients are randomly invited from an ongoing research registry for participation in the trial (Figure 1). The ongoing research registry is made up of all patients who have signed the general research consent (GRC) of the University Hospital Basel.

Based on past-data of the Division of Medical Oncology, around 240-300 patients would be eligible for the trial over a period of 18 months. However, for logistic reasons and capacity of the Department of Pathology, we limit the sample size to 150 participants, with a maximum of three patients selected for invitation to the trial per week.

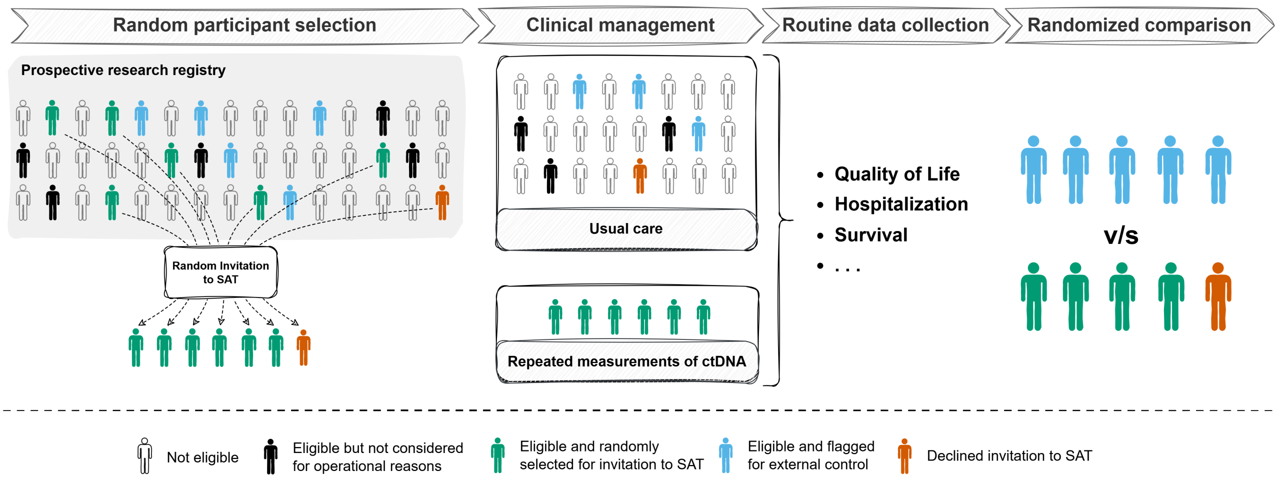


Figure 1: Patients are randomly selected for invitation to the single-arm trial (SAT) from a prospective research registry. Some eligible participants might not be considered for selection for operational reason, e.g., inpatients who require emergency treatment before trial staff was made aware of their existence. All outcome data are obtained from routinely collected data. The design enables an unconfounded (randomized) comparison of patients who were selected for invitation to the SAT with patients who were not selected for invitation.

4.1.1 Trial population

4.1.1.1 Inclusion criteria

1. All adult patients who are part of the University Hospital Basel prospective research registry, i.e., have signed the GRC, with
2. A proven advanced solid malignant disease, and
3. An indication for medical anti-cancer treatment (including combined chemo-radiotherapy or immuno-radiotherapy).

4.1.1.2 Exclusion criteria

1. Patients with primary brain tumors,
2. Patients with primary resectable disease, and
3. Patients with prior treatment for advanced or metastatic disease.

4.1.2 Random Invitation to the SAT

4.1.2.1 Screening list

We build a weekly screening list containing all patients who have an appointment for a first consultation at the Division of Medical Oncology in the following week in REDCap (v. 14.3.14), using routinely collected data from the electronic patient records. The list is randomly ordered by sorting a hashed version (SHA-256) of the sequentially generated patient hospital ID. The trial team (at least one board-certified oncologist) checks whether patients fulfill the eligibility criteria every Friday afternoon.

4.1.2.2 Invitation to SAT

Descending the screening list row by row, we select up to three patients per week who will be invited to the trial. The decision to invite is made randomly, using a random sampling table uploaded to REDCap. The random sampling table was generated by a trialist not associated with the trial using R version 4.3.1 (2023-06-16) R Core Team [1]. The table was stratified by presence or absence of a primary diagnosis of a lung tumor, with a block size of 9, and has odds of 2:1 for an invitation to the trial. All investigators are blinded to the random sampling table.

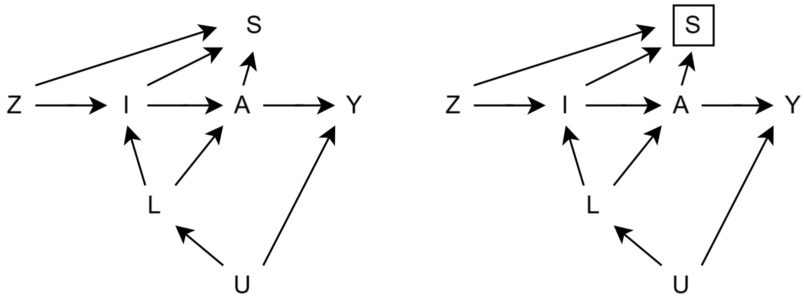
Once three patients have been randomly selected for an invitation to the trial, the list is closed for the current week and a new one is started the next week. Inpatients can also be invited when fulfilling eligibility criteria and slots are available for the given week.

4.1.3 Randomized comparison with external comparator

Randomly inviting patients from a prospective research registry to the SAT enables a randomized comparison with patients who were not invited to the trial, as the data on outcomes for the latter can be obtained from routine data. As visualized in Figure 2, invitation to the trial and the outcomes of interest have no common causes, because the decision to select a patient for invitation to the SAT was made at random.

As all patients who are considered for LIQPLAT are part of the prospective research registry and have therefore agreed for their data to be used for research purposes, we can compare the outcomes of patients who were invited to the trial to those who were not invited. However, a patient's decision to accept the invitation to the SAT is not random and likely to be influenced by factors (U) which also cause the outcome. In other words, the patients who were selected for invitation to the SAT and patients who were not selected for invitation to the SAT are exchangeable regarding their potential outcomes [2], but participants of the SAT are not exchangeable with non-participants. This means that for the randomized comparison, the random selection for invitation to the SAT is considered as the treatment.

Even though LIQPLAT is not strictly speaking a randomized controlled trial (RCT), when considering selection for invitation to the trial as treatment, it can be analyzed analogously to an RCT. We will therefore specify and analyze comparative analyses in line with the estimand framework [3], where we consider invitation to the SAT as the treatment.



(a) Dag 1 (b) Dag 2

Figure 2: Simplified causal diagram for the LIQPLAT trial with random selection for invitation to the SAT as Z, actual invitation I, acceptance of the invitation A, and outcome Y. U represents the unmeasured common causes of I, A and Y. S is an indicator of selection for into the population who accepted the invitation . The Z-Y association in the population who accepted the invitation (a restriction represented by the box around S in the second causal diagram) will be affected by selection bias unless all prognostic factors L are correctly identified and adjusted for.

4.1.4 Outcome data

All participants of LIQPLAT are part of the University Hospital's prospective research registry and have thus consented to further use of their routinely collected data for research purposes. We will obtain outcome data almost exclusively from routinely collected data from electronic health records (EHR), which are mirrored in the clinical data warehouse (CDWH), and REDCap databases of the Division of Medical Oncology.

Trial specific data, such as dates of random selection, are stored within REDCap database specific to LIQPLAT.

4.2 Primary Endpoints for Randomized Comparison

This SAP focuses on the three primary endpoints for the randomized comparison between the group selected for invitation to the SAT and the external control group (not selected for invitation).

1. **Overall Survival (OS):** Time from date of random selection until death due to any cause.
2. **Quality of Life (QoL):** Patient-reported global quality of life measured longitudinally using EORTC questionnaires during routine care.
3. **Time Alive and Out of Hospital (TAOOH):** Time spent alive and not hospitalized (including emergency room visits) during the follow-up period.

Secondary endpoints as described in the protocol (e.g., PFS, number of imaging exams) will be analyzed but are considered more exploratory.

4.3 Analysis Populations

4.3.1 Intention-to-Treat (ITT) Population

The ITT population includes all patients from the prospective research registry who were eligible for LIQPLAT according to the criteria in Section 4.1.1. It comprises two groups defined by the random selection process:

1. **Intervention Group (Selected for Invitation):** All eligible patients randomly selected for invitation to participate in the LIQPLAT single-arm trial (SAT).
2. **External Control Group (Not Selected for Invitation):** All eligible patients who were considered for invitation but not randomly selected during the same recruitment period.

This population forms the basis for the primary analyses of Overall Survival and Time Out of Hospital.

4.3.2 Principal Stratum Population (for QoL Analysis)

The primary analysis for the Quality of Life endpoint will be conducted on the **Principal Stratum Population**. This population is defined as the subset of ITT patients who would have survived the entire 26-week (6-month) follow-up period, regardless of their assignment to be invited to the SAT or not. In practice, this population will be approximated by including all patients from the ITT population who were observed to be alive at the end of the 26-week follow-up. The rationale for restricting the QoL analysis to this Principal Stratum is detailed in Section 7.2.3.3.

5 General Statistical Considerations

5.1 Framework

All primary analyses will be conducted within a Bayesian framework. The inferential goal is to estimate the full posterior distribution for the primary estimand for each outcome. This allows for direct probabilistic statements about the plausible magnitude and direction of the treatment effects. We will report key summaries of the posterior distributions (e.g., median, 95% credible interval) and relevant probabilities (e.g., probability of benefit, $P(\tau > 0)$ or $P(\tau < 0)$ depending on endpoint direction). We will not pre-specify thresholds for dichotomous claims of success or failure based on these probabilities.

5.2 Missing Data Handling

Missing data are expected for baseline covariates and longitudinal outcomes (QoL states, TAOOH states). Death is unlikely to be missing, as the data will be obtained from a very complete registry.

Multiple Imputation by Chained Equations (MICE) will be used, assuming data are Missing At Random (MAR). Imputations will be run using a maximum of 50 iterations for each imputed dataset. We will create $m = 50$ completed datasets. For QoL[1] and TAOOH[2] we will use multilevel imputation, with the patient

[1]More details on multiple imputation strategy for the QoL outcome can be found here. The variables used for imputation are: Age at enrollment, patient sex, ECOG performance status, primary cancer diagnosis, serum albumin, C-reactive protein (CRP), total days in hospital or emergency room, number of blood product units received, indicator for starting next line of treatment, indicator for disease progression, death indicator, Nelson-Aalen cumulative hazard estimate, treatment group allocation, day of questionnaire completion, and EORTC QLQ-C30 question 30.

identifier as the cluster variable using Predictive Mean Matching (PMM) only. For OS[3], data will be imputed in wide format. Convergence will be assessed via trace plots, the plausibility of the imputations will assess using diagnostics graphs comparing the distribution of observed and imputed data.

Each model (OS, QoL, TAOOH) will be fitted to each of the $m = 50$ imputed datasets. The posterior draws for all parameters and derived estimands from the 50 analyses will be combined (stacked) to form the final posterior distribution for reliable inference [4].

5.3 Software

Analyses will be performed using R (version 4.4.3 or later) [1]. Bayesian models will be fitted using Stan via the `rstanarm` [5] and `rmsb` [6] packages. Multiple imputation will use the `mice` [7] and `miceadds` [8] packages. For data wrangling and plotting we use the `tidyverse` [9]. To assess the convergence of the Markov Chain Monte Carlo (MCMC), we will check $\hat{R}$ for all parameters, with a target value < 1.01, and inspect trace plots to confirm adequate mixing of the chains.

6 Sample Size

The trial aims to enroll 150 participants who accept the invitation to the single-arm trial (SAT). This target sample size is driven by feasibility, including logistical constraints (such as a maximum of three invitations per week) and the capacity of the Department of Pathology. Based on estimates of patient acceptance, we anticipate needing to invite approximately 180 eligible patients to achieve this target (an acceptance rate of ≈ 83).

The random invitation process, as described in Section 4.1.1, uses a $2 : 1$ allocation ratio (invitation vs. no invitation). Applying this ratio to the 180 invited patients implies an expected external control group (not selected for invitation) of approximately 90 patients.

Therefore, the total Intention-to-Treat (ITT) population for the randomized comparison is expected to be $N = 270$ (180 patients in the intervention group [selected for invitation] and 90 patients in the external control group). All simulation studies for power and operating characteristics are based on this total sample size of $N = 270$.

This trial is not formally powered for a single primary endpoint using a frequentist framework, as the sample size is fixed by feasibility. Instead, the Bayesian analyses will provide posterior distributions and probabilities of benefit for the three primary endpoints. The operating characteristics presented in the analysis sections provide an indication of the study's ability to detect various plausible effect sizes given this fixed sample size.

It is crucial to note that the primary estimand for all outcomes is the effect of the treatment policy, which compares the group selected for invitation ($A = 1$) to the group not selected for invitation ($A = 0$). This treatment policy estimand is inherently diluted by non-adherence (i.e., the ≈ 30 patients in the $A = 1$ group who are expected to decline participation).

The simulation studies (e.g., for TAOOH, Table 2) did not explicitly model this non-uptake mechanism. Instead, the true effect sizes used in the simulations (e.g., a true OR of 0.7) are assumed to represent this

[2] More details on multiple imputation strategy for the TOAAH outcome can be found here. The variables used for imputation are: age at enrollment, patient sex, ECOG performance status, and primary cancer diagnosis, serum albumin, C-reactive protein (CRP), serum bilirubin, and lactate dehydrogenase (LDH), number of blood product units received, indicator for starting next line of new treatment and indicator for disease progression, death indicator, Nelson-Aalen cumulative hazard estimate, and treatment group allocation, and day after random selection, indicator for being in hospital, and indicator for emergency room visit.

[3] More details on multiple imputation strategy for the OS outcome can be found here. The variables used for imputation are: age at enrollment, patient sex, ECOG performance status, and primary cancer diagnosis, serum albumin, C-reactive protein (CRP), serum bilirubin, and lactate dehydrogenase (LDH), total days in hospital or emergency room and number of blood product units received, indicator for starting next line of new treatment and indicator for disease progression, and death indicator, Nelson-Aalen cumulative hazard estimate, and treatment group allocation.

already-diluted ITT effect. In other words, a simulated OR of 0.7 reflects the average effect of the invitation strategy across all 180 invited patients (including non-acceptors), compared to the 90 non-invited patients.

7 Analysis

7.1 Overall Survival (OS)

7.1.1 Estimand Definition

Attribute	Definition
Target Population	Adult (≥ 18 years) patients with advanced solid cancer, who have not yet started first line therapy.
Treatment Arms	(A=1): Selection for invitation to ctDNA-guided care (SAT).
	(A=0): No selection for invitation (External Control Group / SoC).
Variable	Time-to-event: Time (in days) from random selection to death from any cause.
Time Horizon	$t_{max} = 182$ days (6 months).
Summary Measure	Difference in Restricted Mean Survival Time (RMST) at $t_{max} = 182$ days.
Intercurrent Events	See Section 7.1.2.

7.1.1.1 Formal Definition

We use the potential outcomes framework [2]. Let T_i^a be the potential time-to-death for individual i under treatment assignment $a \in \{0, 1\}$. The survival function under treatment a is $S^a(t) = P(T^a > t)$.

The Restricted Mean Survival Time (RMST) under treatment a up to the time horizon $t_{max} = 182$ days is the area under the survival curve:

$$\text{RMST}^a(t_{max}) = \int_0^{t_{max}} S^a(t) dt$$

Our primary estimand for OS is the Average Treatment Effect (ATE) on the RMST at 182 days, denoted τ_{OS}:

$$\tau_{OS} = \text{RMST}^1(t_{max}) - \text{RMST}^0(t_{max})$$

τ_{OS} represents the average difference in days lived within the first 182 days comparing the group selected for invitation to the SAT versus the external control group.

7.1.2 Handling of Intercurrent Events (OS)

Intercurrent Event	Strategy	Rationale
SAT not offered after selection	Treatment Policy	Analyze as selected (part of A=1). Preserves ITT principle; estimates effect of the *strategy* of offering ctDNA-guided care.
Patient declined participation in SAT	Treatment Policy	Analyze as selected (part of A=1). As above.
Patient changes hospital / Lost to F/U	Censoring (if alive)	Death information is obtained via CDWH / administrative databases, minimizing informative censoring due to death ascertainment. If lost to follow-up,

Intercurrent Event	Strategy	Rationale
		e.g., moved out of country, and known alive at last contact, standard survival censoring applies.
Control patient receives ctDNA test	Treatment Policy	Analyze as not selected (part of A=0). Reflects real-world practice where ctDNA may be used outside the trial context.
Discontinuation of ctDNA monitoring (SAT)	Treatment Policy	Analyze as selected (part of A=1). Adherence does not affect assignment group.

7.1.3 Analysis Method (OS)

7.1.3.1 Statistical Model

The analysis model is a Bayesian proportional hazards survival model. It is defined by the hazard rate $h_i(t)$ for each individual i at time t.

$$
\begin{aligned}
\mathrm{day}_i, \mathrm{death}_i &\sim \text{Survival }(h_i(t)) \\
h_i(t) &= h_0(t) \cdot \exp(\eta_i) \\
\eta_i &= \beta_0 + \beta_{tx} \cdot \mathrm{tx}_i + \sum_{j=2}^{3} \beta_{ecog,j} \cdot \mathrm{ecog}_{ij} + \sum_{k=2}^{5} \beta_{diag,k} \cdot \mathrm{diag}_{ik} \\
&\quad + f_1(\mathrm{crp}_i) + f_2(\mathrm{albumin}_i) \\
h_0(t) &= \sum_{k=1}^{5} \gamma_k \cdot M_k(t) \\
&\quad \text{where :} \\
&\quad\quad f_1 = \sum_{m=1}^{2} \beta_{crp,m} \cdot \mathrm{B}_{m,crp}(\mathrm{crp}_i) \\
&\quad\quad f_2 = \sum_{n=1}^{2} \beta_{alb,n} \cdot \mathrm{B}_{n,alb}(\mathrm{albumin}_i) \\
\beta_0 &\sim \mathcal{N}(0, 20) \\
\beta_{tx} &\sim \mathcal{N}(0, 0.55) \\
\beta_{\mathrm{ecog}, j \in 2,3} &\sim \mathcal{N}(0, 2.5) \\
\beta_{\mathrm{diag}, k \in 2..5} &\sim \mathcal{N}(0, 2.5) \\
\beta_{crp, m \in 1,2} &\sim \mathcal{N}(0, 2.5) \\
\beta_{alb, n \in 1,2} &\sim \mathcal{N}(0, 2.5) \\
\gamma &\sim \text{Dirichlet }(\mathbf{1}_5)
\end{aligned}
$$

We assume the observed data for individual i, which consists of a time-to-event indicator (day), and an event indicator (death) follows a survival process which is defined by the hazard function $h_i(t)$.

The hazard rate $h_i(t)$ for an individual is the product of a flexible baseline hazard, $h_0(t)$, and the exponentiated effect of their covariates, $\exp(\eta_i)$.

- **Baseline Hazard**: $h_0(t)$ is modeled using M-splines with 5 degrees of freedom (three internal knots). $M_k(t)$ are non-negative M-spline basis functions.[4]
- **Linear Predictor**: η_i contains the intercept (β_0), the effect of treatment ($\beta_{tx} \cdot \mathrm{tx}_i$), the effects of the categorical predictors (represented by sums over their dummy variables), and the effects of the continuous predictors. We model the prognostic effects of the continuous baseline covariates using restricted cubic splines with three knots (boundary knots at the minimum and maximum, inner knot at the median). This generates two basis functions (B_m and B_n) for each covariate. The model thus estimates $\beta_{crp,1}$, $\beta_{crp,2}$ and $\beta_{alb,1}$, $\beta_{alb,2}$.[5] In historical data this approach led to more precise estimates of the baseline hazards than a simple log-transformation.
- **Priors**: The model specifies Normal priors for all regression coefficients. For treatment (β_{tx}), we use a Normal(0, 0.55) prior. For all other coefficients we use Normal(0, 2.5) priors. A Dirichlet(1) prior is used for the M-spline coefficients γ. The Dirichlet prior is used because the spline coefficients are constrained by `rstanarm` to be positive and sum to 1 for identification. The Normal(0, 0.55) prior on the treatment parameter places approximately 80% probability mass on Hazard Ratios between 0.5 and 2.0, reflecting a belief that the intervention is unlikely to more than halve or double the hazard rate.

7.1.3.2 Implementation

The model will be fitted using the `stan_surv` function from the `rstanarm` package in R. Convergence will be assessed using $\hat{R}$ and trace plots.

```r
# Define priors
tx_scale <- 0.55
default_scale <- 2.5
my_prior <- normal(
  location = c(rep(0, 12)),
  scale = c(tx_scale, rep(default_scale, 11))
)


# Fit the model
model_os <- stan_surv(
    formula = Surv(day, death) ~ tx + ecog_fstcnt + diagnosis + ns(albumin, df = 2) +
ns(c_reactive_protein, df = 2),
    data = data,
    basehaz = "ms",
    basehaz_ops = list(df = 5),
    prior = my_prior,
    prior_intercept = normal(0, 20),
    prior_aux = dirichlet(1),
    chains = 4,
    cores = 4,
    seed = 68818,
    iter = 4000
)
```

[4] In historical data, we compared the following hazard functions: exponential, weibull, B-splines with 5 degrees of freedom (df), M-splines with 5 df, and M-splines with 10 df. The predictive performance of these models according to LOOCV did not differ substantially. We therefore chose to model the hazard using M-splines with 5 df, for more flexibility should the trial data be substantially different, but more parsimony than when using 10 df. More details can be found here.

[5] While we refer to these as restricted cubic splines, the model was fit using the natural spline function, `ns()`, from the R `splines` package. This function implements the same model (a cubic spline restricted to be linear in the tails) but uses a B-spline basis. We chose to do this, as the truncated power basis implementation as used in `rcs()` from the R `Hmisc` package created convergence issues.

7.1.3.3 Obtaining the Estimand

1. **Generate Posterior Survival Curves:** Using the fitted model (`model_os`), generate posterior draws of the standardized survival curves for each treatment group ($a = 0$ and $a = 1$) up to $t_{max} = 182$ days. This involves predicting survival probabilities for each individual under both treatment scenarios and averaging these predictions over the sample's covariate distribution for each posterior draw. The `posterior_survfit` function in `rstanarm` with `standardise = TRUE` will be used.
2. **Calculate RMST per Draw:** For each posterior draw, calculate the RMST for both treatment groups (RMST^0 and RMST^1) by integrating the respective survival curves from $t = 0$ to $t_{max} = 182$ days. We will approximate the integral using the trapezoidal rule on the discrete time points generated by `posterior_survfit`.
3. **Calculate RMST Difference per Draw:** For each posterior draw, compute the difference $\tau_{OS,\text{ draw}} = \text{RMST}^1_{\text{draw}} - \text{RMST}^0_{\text{draw}}$.
4. **Summarize Posterior:** The collection of $\tau_{OS,\text{ draw}}$ values forms the posterior distribution of the primary estimand. Summarize this distribution using the posterior median and a 95% credible interval.

```r
# --- Calculate RMST ---
# Get posterior survival curves for tx=0
ps_0_list <- posterior_survfit(
    model,
    newdata = data_for_model |> mutate(tx = 0),
    times = 0,
    extrapolate = TRUE,
    standardise = TRUE,
    control = list(edist = 182, epoints = 200),
    return_matrix = TRUE,
    draws = 4000
)

std_surv_matrix_0 <- do.call(cbind, ps_0_list)
time_points <- sapply(ps_0_list, attr, "times")

# Get posterior survival curves for tx=1
ps_1_list <- posterior_survfit(
    model,
    newdata = data_for_model |> mutate(tx = 1),
    times = 0,
    extrapolate = TRUE,
    standardise = TRUE,
    control = list(edist = 182, epoints = 200),
    return_matrix = TRUE,
    draws = 4000
)
std_surv_matrix_1 <- do.call(cbind, ps_1_list)

# Calculate RMST for tx=0 using trapezoidal rule
delta_t <- diff(time_points)
s_t1_0 <- std_surv_matrix_0[, -ncol(std_surv_matrix_0)]
s_t2_0 <- std_surv_matrix_0[, -1]
trapezoid_areas_0 <- 0.5 * (s_t1_0 + s_t2_0) * matrix(delta_t, nrow = nrow(s_t1_0),
ncol = length(delta_t), byrow = TRUE)
rmst_0 <- rowSums(trapezoid_areas_0)

# Calculate RMST for tx=1
```

```r
s_t1_1 <- std_surv_matrix_1[, -ncol(std_surv_matrix_1)]
s_t2_1 <- std_surv_matrix_1[, -1]
trapezoid_areas_1 <- 0.5 * (s_t1_1 + s_t2_1) * matrix(delta_t, nrow = nrow(s_t1_1),
ncol = length(delta_t), byrow = TRUE)
rmst_1 <- rowSums(trapezoid_areas_1)

# Calculate the difference
rmst_diff <- rmst_1 - rmst_0

# Combine and save results
rmst_results <- data.frame(rmst_0, rmst_1, rmst_diff)
```

7.1.3.4 Simulated Power (OS)

[Still need to finalize the simulations].

7.2 Quality of Life (QoL)

7.2.1 Data

In LIQPLAT, Quality of Life (QoL), measured using the European Organisation for Research and Treatment of Cancer (EORTC) QLQ-C30 and QLQ-C15 questionnaires [10].[6] The QoL scale (question 30) is an ordinal outcome with seven levels, ranging from 1 (very poor) to 7 (excellent), as shown below:

How would you rate your overall quality of life during the past week?

1	2	3	4	5	6	7
Very poor						Excellent

For modeling purposes, we invert the scale. We will also assume that the categories that are not labeled according to EORTC can be labeled as follows:[7]

1	2	3	4	5	6	7
Excellent	Very Good	Good	Average	Below Average	Poor	Very poor

In the LIQPLAT, QoL questionnaires are completed at clinical appointments, which occur frequently but at irregular intervals for each patient. We anticipate an average interval of about six weeks. This data structure has two key implications.

1. **Time-Dependent Correlation:** A patient's QoL at one time point is correlated with their QoL at preceding time points. This within-subject correlation should be accounted for [13]. In other words, multiple measurements within a subject are *not* exchangeable, as would be assumed by a simple random intercept and the associated compound symmetric correlation structure [13].
2. **Irregular Gaps:** The correlation between two measurements likely weakens as the time gap between them increases. The analytical model should explicitly incorporate the time gap between consecutive measurements to better approximate the decaying correlation structure. If we assume that the irregularity

[6] The EORTC's scoring manual advocates for treating the responses as numeric [11], which we will not do. The shortcomings of treating ordinal scores as if they were numeric have explained in detail elsewhere [12].

[7] We acknowledge that not everyone would assign precisely these labels, but believe them to be agreeable to the majority of patients. The labels will serve for easier communication of results.

of follow-up is random conditional on a set of baseline covariates and the previous QoL state, the irregular spacing could be used to reconstruct QoL trajectories over the entire follow-up period.

3. **Competing Risk:** In a population with advanced cancer, death is a frequent outcome that prevents any further QoL measurement. This represents a competing risk. As detailed in Section 7.2.3.3 this competing risk is a major challenge.

7.2.2 General Approach

We propose to make use of the irregular, near-random follow-up of usual care and its QoL measurements to reconstruct the quality of life trajectories for the entire duration of follow-up (see Figure 3).

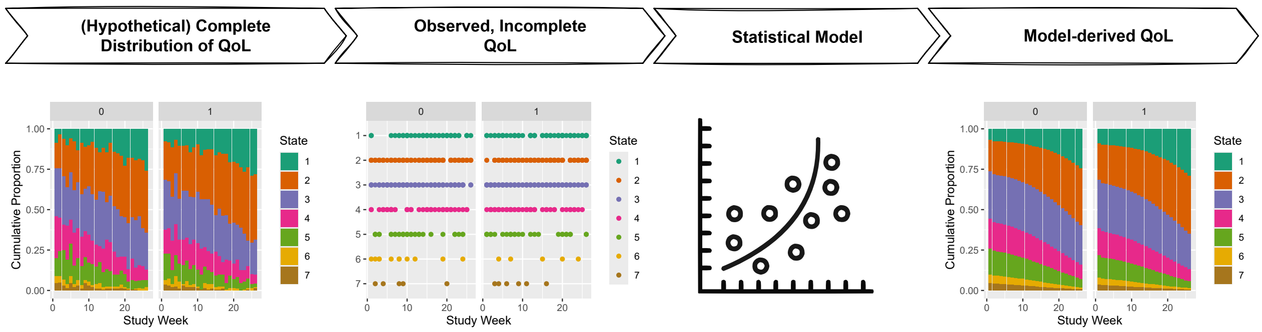


Figure 3: Conceptual overview of the statistical approach for reconstructing Quality of Life (QoL) trajectories. The analysis bridges the gap between the hypothetical, complete QoL data (far left), from which empirical state occupancy probabilities (SOPs) could be directly calculated, and the actual observed data, which is sparse and irregularly spaced (second from left). A statistical model (third from left) is fitted to these sparse observations to generate complete, model-derived SOPs for the entire follow-up period (far right), reconstructing the full QoL trajectory for the population and providing associated uncertainty estimates.

7.2.3 Estimand Definition (QoL)

Attribute	Definition
Target Population	The Principal Stratum of adult (≥ 18 years) patients with advanced solid cancer, who have not yet started first-line therapy, defined as patients who would survive at least 26 weeks after their first appointment.
Treatment Arms	(A=1): Selection for invitation to ctDNA-guided care (SAT).
	(A=0): No selection for invitation (External Control Group / SoC).
Variable	Ordinal QoL state (7 levels, derived from EORTC QLQ-C30 Q30) measured longitudinally at irregular intervals. We use an inverted scale where 1='Excellent', …, 7='Very Poor'.
Time Horizon	$t_{max} = 26$ weeks (6 months).
Summary Measure	Difference in mean number of weeks spent with "Good" QoL (defined as states 1, 2, or 3) during the 26-week period, within the Principal Stratum.
Intercurrent Events	See Section 7.2.4.

7.2.3.1 Formal Definition (QoL)

Let S_i^a be an indicator for survival of patient i at 26 weeks under treatment assignment $a \in \{0, 1\}$. $S_i^a = 1$ if patient i survives, $S_i^a = 0$ otherwise. The Principal Stratum of "always-survivors" is the sub-population of patients for whom $S_i^1 = S_i^0 = 1$.

Let Y_{it}^{a} be the potential ordinal QoL state (1-7, 1=best) for patient i at week t under treatment a. Let W_i^a be the potential total number of weeks patient i spends in a "Good" QoL state ($j \in \{1, 2, 3\}$) under treatment a over 26 weeks:

$$W_i^a = \sum_{t=1}^{26} \sum_{j=1}^{3} \mathbb{I}(Y_{it}^a = j)$$

The primary estimand for QoL, the Principal Stratum Average Treatment Effect (psATE) on weeks in a "Good" state, denoted τ_W^{psATE}, is:

$$\tau_W^{\text{psATE}} = \mathbb{E}[W_i^1 \mid S_i^1 = 1, S_i^0 = 1] - \mathbb{E}[W_i^0 \mid S_i^1 = 1, S_i^0 = 1]$$

7.2.3.2 Identification Assumption (QoL)

Identification of τ_W^{psATE} by analyzing only those patients *observed* to survive 26 weeks relies on the key assumption that selection for invitation to the SAT has no effect on 26-week survival status: For all patients i, $S_i^1 = S_i^0$. Under this assumption, the group observed to survive is equivalent to the "always-survivor" Principal Stratum.

7.2.3.3 Rationale for Principal Stratum (QoL)

Modeling the QoL trajectory while incorporating the absorbing state of death proved problematic with sparse intermediate QoL measurements common in routine care. Our simulations indicated that interval-censoring approaches for missing QoL states led to biased estimates, particularly overestimating the probability of death (Section 11.1). Assuming no treatment effect on 6-month survival allows us to focus the QoL analysis on the stratum of patients who survive this period regardless of treatment, providing a potentially less biased estimate of the QoL effect *for this subgroup*, conditional on the validity of the no-survival-effect assumption.

7.2.4 Handling of Intercurrent Events (QoL)

Intercurrent Event	Strategy	Rationale
Death	*Principal Stratum*	Death before 26 weeks defines exclusion from the primary analysis population. This relies on the assumption in Section 7.2.3.2.
Discontinuation of ctDNA monitoring (SAT)	*Treatment Policy*	Analyze as selected (part of A=1). Preserves ITT principle; estimates the effect of the *strategy* of offering ctDNA-guided care, irrespective of adherence.
Switch to best supportive care (BSC)	*???*	Patient remains in their assigned group (A=1 or A=0). BSC could lead to cessation of QoL data collection.
Treatment at another hospital / LFU	*???*	???
SAT not offered after selection	*Treatment Policy*	Analyze as selected (part of A=1).
Patient declined participation in SAT	*Treatment Policy*	Analyze as selected (part of A=1).
Control patient receives ctDNA test	*Treatment Policy*	Analyze as not selected (part of A=0).

⚠ Warning

Challenge: Informative Missingness due to Routine Care Data Collection A major limitation is that QoL is measured during routine appointments. If the ctDNA strategy influences clinical decisions (e.g., earlier switch to BSC, different follow-up intensity), it affects *both* the patient's true QoL *and* the probability of observing QoL. This creates a potential for informative missingness. See Section 8.1.

7.2.5 Analysis Method (QoL)

To target the estimand we use a first-order Markov longitudinal ordinal transition model [14]. This model analyzes the probability of transitioning between QoL states from one week to the next, conditional on the previous state and the time gap since the last measurement. This approach accounts for the ordinal nature, longitudinal correlation, irregular measurement times, and provides a framework to derive interpretable summary measures like time spent in specific states. A higher-order Markov model would no be feasible, as quality of life states prior to baseline are not available.

7.2.5.1 Statistical Model: Bayesian First-Order Markov Ordinal Transition Model

Let y_{it} be the ordinal QoL state (1-7, 1=best) for patient i at week t. Let $y_{it'}$ be the last observed state at a prior week t', and let the time gap be $\Delta t = t - t'$. The model is a cumulative logit model for transition probabilities:

$$\operatorname{logit}\left(P(y_{it} \geq j \mid y_{it'})\right) = \alpha_j + \eta_{it} + \gamma_{it,j} \qquad \text{for } \; j = 2, \ldots, 7$$

$$\begin{aligned}
\eta_{it} &= \beta_{\text{tx}} \cdot \text{tx}_i + f(t) + \sum_{k=2}^{7} \beta_{\text{yprev }=k} \cdot \mathbb{I}(y_{it'} = k) \\
&\quad + \beta_{\text{gap}} \cdot \Delta t + \sum_{k=2}^{7} \beta_{\text{yprev }=k \times \text{ gap}} \cdot \mathbb{I}(y_{it'} = k) \cdot \Delta t \\
&\quad + \sum_{j=2}^{3} \beta_{ecog,j} \cdot \text{ecog}_{ij} + \sum_{k=2}^{5} \beta_{diag,k} \cdot \text{diag}_{ik} \\
\gamma_{it,j} &= (\tau \cdot t) \cdot j
\end{aligned}$$

where:

$$f(t) = \sum_{p=1}^{3} \beta_{t,p} \cdot B_p(t)$$

$$\begin{aligned}
\pi_1, \ldots, \pi_K &\sim \text{Dirichlet}\ (0.308)\ , \text{ which induces priors on } \ \alpha_j \\
\beta_{tx} &\sim \mathcal{N}(0, 0.940) \\
\beta_{t,p \in 1,2,3} &\sim \mathcal{N}(0, 100) \\
\beta_{gap} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{yprev }=k \in 2..7} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{yprev }=k \times \text{ gap } \in 2..7} &\sim \mathcal{N}(0, 100) \\
\beta_{ecog, j \in 2,3} &\sim \mathcal{N}(0, 100) \\
\beta_{diag, k \in 2..5} &\sim \mathcal{N}(0, 100) \\
\tau &\sim \mathcal{N}(0, 100)
\end{aligned}$$

Where:

- α_j: Category-specific intercepts (cutpoints).
- η_{it}: Linear predictor for effects assuming proportional odds (PO).
  - β_{tx}: Main effect of treatment selection (A=1 vs A=0).[8]
  - $f(\text{week}_t)$: Flexible function of study week[9] modeled as a restricted cubic spline with 4 knots.[10]
  - $\beta_{\text{yprev }=k}$: Effect of the previous QoL state (categorical, state 1 as reference).[11]
  - β_{gap}: Linear effect of the time gap Δt.[12]
  - $\beta_{\text{yprev }=k \times \text{ gap}}$: Interaction between previous state and time gap.
  - $\beta_{ecog,j \in 2,3}$, $\beta_{diag,k \in 2..5}$: Effects of baseline covariates (ECOG status, diagnosis category).[13]
- $\gamma_{it,j}$: Models deviation from PO for the effect of time. We allow the effect of time to be non-proportional, constrained to be linear in the outcome category j. τ is the non-proportional odds parameter for time.

7.2.5.2 Prior Specifications

- **Intercepts (α_j)**: Priors induced by a Dirichlet(0.308)[14] distribution on baseline cell probabilities. Ensures ordering $\alpha_2 < ... < \alpha_7$.[15]
- **Treatment Parameter (β_{tx})**: We use prior predictive simulation to select a prior for β_{tx} that implies a clinically plausible distribution for our estimand τ_W^{psATE}. Our simulation, based on historical data, suggests that a prior of $\mathcal{N}(0, 0.940)$ on β_{tx} produces a prior a prior distribution for τ_W^{psATE} with a median at 0 weeks that has 80% of its prior mass between $[-6.0, +4.0]$.
- **Other Coefficients**: Non-informative priors, $\mathcal{N}(0, 100)$, (Default priors in `rmsb`).

[8] A treatment effect, if present, is unlikely to be constant over time. CtDNA sampling can take weeks and is less likely to change first-line than second-line therapy of participants. An effect in the first weeks after baseline is thus very unlikely. However, given our sample size, we are likely not able to identify time-varying treatment effects reliably and choose a constant treatment effect for slightly smaller variance in our estimator.

[9] The analysis discretizes time into weekly intervals rather than daily intervals for several pragmatic and methodological reasons. First, in the EORTC questionnaire patients are specifically asked about their quality of life in the last week. Second, given the high proportion of missing data inherent to routine care follow-up, daily estimates could imply a level of precision that is not supported by the data. Third, in LIQPLAT, it is highly unlikely that patients complete more than one questionnaire within a single week, making the week a natural and minimal unit of observation. Finally, this choice significantly reduces the computational burden of the simulation study and the primary analysis (e.g., 26 weekly transitions vs. 182 daily transitions per patient), enhancing the feasibility of the project without a substantial loss of relevant information. The impact of this aggregation on the final estimand is expected to be minimal.

[10] This function is parameterized into $p = 3$ components: one for the linear trend and two for the nonlinear adjustments. This model uses the `rmsb` package, which implements restricted cubic splines using a truncated power basis. This basis is parameterized for convenient interpretation: for $K = 4$ knots, it generates $K - 1 = 3$ basis functions ($B_p(t)$). The first basis function ($B_1(t)$) is the linear term t itself. The subsequent functions ($B_2(t)$,$B_3(t)$) are the nonlinear components derived from truncated cubic functions. Therefore, the coefficients $\beta_{t,1}$, $\beta_{t,2}$, $\beta_{t,3}$ are the weights for the linear component and the two nonlinear components, respectively.

[11] It is unlikely that the effect of the previous state is constant over time. Very poor QoL at the beginning of the study might predict regression to the mean, whereas a very low-quality of life during later stages of the disease might not. However, an interaction of the six-level categorical variable `yprev` with flexibly modeled time would use too many degrees of freedom, given our sample size.

[12] The effect of the previous state is unlikely to decay linearly over time. However, most questionnaires will be repeated after at least some weeks. Even if the decay shortly after a previous answer is non-linear, we anticipate the decay in the expected range of gaps to be approximately linear.

[13]

- Both the functional status (ECOG) at baseline and the cancer diagnosis category are likely to be predictive of quality of life based on our clinical experience and prior research, and are thus included to explain more variation in the outcome. If we observe fewer than five patients with a functional status (ECOG) of 4, we will merge categories 3 and 4 into '3plus', to avoid divergent transitions of the estimate of the effect of ecog 4 because of too few observations. The diagnosis categories will be refactored into a five level categorical variable, with the four most common diagnoses as distinct categories and the rest of diagnoses lumped together as 'other'.

[14] The concentration parameter as calculated by default by the `rmsb` package is $1/(0.8 + 0.35 \cdot \max\ (k, 3))$, which evaluates to 0.308 for a 7-level outcome.

[15] Unlike the TAOOH model, a patient-level random intercept is not included here. With an anticipated average of only ~5 observations per patient for this outcome, estimation of a random intercept variance component is expected to be unstable. While this omission risks Type I error inflation if significant unmodelled correlation exists, it is considered a necessary trade-off given the sparse data.

7.2.5.3 Implementation

The model will be fitted using the `blrm` function from the `rmsb` package in R, restricted to the Principal Stratum population (survivors at 26 weeks). MCMC settings as per OS analysis.

```r
# Example R code for model fitting

library(rmsb)

model_qol <- blrm(
  formula = y ~ tx + rcs(week, 4) + yprev * gap + ecog_fstcnt + diagnosis,
  data = data_survivors,
  ppo = ~week,              # Allow week effect to be non-proportional
  cppo = function(y) y,     # Constraint: non-proportionality linear in y
  iter = 4000, chains = 4, seed = 68818
)
```

7.2.5.4 Obtaining the Estimand (QoL)

1. **Fit Model:** Fit the `model_qol` to each multiply imputed dataset restricted to the Principal Stratum.
2. **Calculate Weekly SOPs:** For each posterior draw from the fitted model, calculate the State Occupancy Probabilities (SOPs), $P(Y_{it}^a = j)$, for each week $t = 1, ..., 26$, for each state $j = 1, ..., 7$, and for each treatment arm $a \in \{0, 1\}$. This involves marginalizing over the distribution of baseline covariates, baseline QoL states, and the Markov process transitions over time.
3. **Calculate Expected Weeks in "Good" State per Draw:** For each posterior draw, calculate the expected number of weeks in a "Good" state (1, 2, or 3) for each treatment arm by summing the relevant SOPs over the 26 weeks:

$$\mathbb{E}[W^a]_{\text{draw}} = \sum_{t=1}^{26} \sum_{j=1}^{3} \text{SOP } (Y_{it}^a = j)_{\text{draw}}$$

4. **Calculate Difference per Draw:** For each posterior draw, compute the difference $\tau_{W,\text{ draw}}^{\text{psATE}} = \mathbb{E}[W^1]_{\text{draw}} - \mathbb{E}[W^0]_{\text{draw}}$.
5. **Summarize Posterior:** The collection of $\tau_{W,\text{ draw}}^{\text{psATE}}$ values forms the posterior distribution of the primary estimand for QoL.

7.2.5.5 Assumptions

! Important

Key assumptions

1. QoL responses can be accurately modeled by using a first-order Markov process. (+)
2. If the effect of time is non-proportional, this can be well approximated using a linear constraint. (+)
3. A quality of life response from a single day is valid for an entire week. (+/-)
4. The treatment effect is proportional, i.e., the odds of changing to a higher category are the same for all thresholds. (+/-)
5. Participants who died during the 6 month follow-up would have died irrespective of their treatment assignment. (+/-)
6. How a previous state affects the current state does not change over time since randomization. (-)
7. The treatment effect, if present, does not change over time. (-)
8. Days where no QoL was assessed are missing at random (MAR). (-)

(+) Indicates that we are confident the assumption is valid, (+/-) indicates that we are uncertain regarding its validity, (-) indicates that we believe the assumption to be strenuous.

Given these assumptions, especially assumptions six to eight, we have **low confidence** in the results. Nevertheless, we believe that other analysis options we have explored are less optimal and would lead us to have very low confidence.

7.2.5.6 Simulation Study Summary (QoL)

Table 1 shows frequentist operating characteristics of the different analyses from our simulation study. The data-generating mechanism assumed an initial sample size of $N = 270$, before conditioning on the Principal Stratum, with an average cumulative incidence of death by 6 months of 18% for both groups, i.e., on average 49 patients were excluded. 85% of post-baseline observations are missing completely at random (MCAR), which equals about 4 post-baseline measurements per patient. Power and Type I error were assessed under a constant true odds ratio (OR) of 0.8 (H_A) and 1.0 (H_0). **[Insert sentence on how many days this equals]**

We will not use the posterior to make a dichotomous claim of benefit. However, for the purpose of evaluating the operating characteristics of our analysis plan, we assess power and Type I error under the hypothetical decision rule of requiring a posterior probability of benefit greater than 0.95. This allows for a standardized comparison of different analytical methods. Details on model specifications, convergence diagnostics, probability of benefit of each iteration can be found in [TBD].

Model	Power	Type I Error
Markov Model	0.450 (0.0157)	0.054 (0.0072)
Random Intercept	0.445 (0.0352)	0.044 (0.0101)
Time-to-Deterioration	0.112 (0.0100)	0.047 (0.0067)
6-Month Ordinal	0.197 (0.0126)	0.050 (0.0069)
Change from Baseline	0.176 (0.0120)	0.050 (0.0069)

Table 1: Bayesian Power and Type I error when $P(\text{benefit}) > 0.95$ is chosen as a cut-off for benefit for the five endpoints (Monte Carlo SE in parenthesis).

7.3 Time Out of Hospital (TAOOH)

7.3.1 Estimand Definition (TAOOH)

Attribute	Definition
Target Population	Adult (≥ 18 years) patients with advanced solid cancer, who have not yet started anti-cancer therapy.
Treatment Arms	(A=1): Selection for invitation to ctDNA-guided care (SAT).
	(A=0): No selection for invitation (External Control Group / SoC).
Variable	Weekly health state: 1=Alive/Out-of-Hospital, 2=Alive/In-Hospital(Planned), 3=Alive/In-Hospital(ER), 4=Alive/In-Hospital(Unplanned), 5=Dead.
Time Horizon	$t_{max} = 26$ weeks (6 months).
Summary Measure	Difference in mean number of weeks spent Alive and Out of Hospital (State 1) during the 26-week period.
Intercurrent Events	Primarily death, which is modeled as the absorbing state (State 5). See Section 7.1.2 for considerations.

7.3.1.1 Formal Definition (TAOOH)

Let Y_{it}^a be the potential health state (1-5) for individual i at week t under treatment assignment $a \in \{0, 1\}$. State 1 represents "Alive and Out of Hospital".

Let W_i^a be the potential total number of weeks individual i spends alive and out of the hospital (State 1) under treatment a over the 26-week period:

$$W_i^a = \sum_{t=1}^{26} \mathbb{I}(Y_{it}^a = 1)$$

Our primary estimand for TAOOH is the Average Treatment Effect (ATE) on this outcome, denoted τ_{TAOOH}:

$$\tau_{TAOOH} = \mathbb{E}[W_i^1] - \mathbb{E}[W_i^0]$$

τ_{TAOOH} represents the population-average difference in weeks spent alive and out of the hospital over 26 weeks, comparing the group selected for invitation versus the external control group.

7.3.2 Handling of Intercurrent Events (TAOOH)

Death is explicitly modeled as the absorbing state (State 5) within the transition model. Other intercurrent events follow the Treatment Policy approach as outlined for OS (Section 7.1.2).

[**MORE DETAILS**]

7.3.3 Analysis Method (TAOOH)

7.3.4 Rationale for Ordinal Longitudinal Modeling (TAOOH)

We define a patient's state weekly based on their location and vital status: 1=Alive and Out of Hospital, 2=Alive and In Hospital (Planned Admission), 3=Alive and In Hospital (ER), 4=Alive and In Hospice (Unplanned Admission), 5=Dead. This creates a longitudinal ordinal outcome (with state 5 being absorbing).

7.3.5 Statistical Model: Bayesian Second-Order Markov Ordinal Transition Model with Random Effects

Let y_{it} be the ordinal health state (1-5) for patient i at week t. Let $y_{i,t-1}$ and $y_{i,t-2}$ be the states at the previous two weeks. The model is a cumulative logit model:

$$
\begin{aligned}
\text{logit}\ \big(P\big(y_{it} \geq j \mid y_{i,t-1}, y_{i,t-2}\big)\big) &= \alpha_j + \eta_{it} + \gamma_{0i} && \text{for} \quad j = 2, ..., 5 \\
\eta_{it} &= \beta_{\text{tx}} \cdot \text{tx}_i + f(t) \\
&\quad + \sum_{k=2}^{4} \beta_{\text{yprev1},k} \cdot \mathbb{I}\big(y_{i,t-1} = k\big) + \sum_{l=2}^{4} \beta_{\text{yprev2},l} \cdot \mathbb{I}\big(y_{i,t-2} = l\big) \\
&\quad + \sum_{k=2}^{4} \beta_{t \times yprev1,k} \cdot t \cdot \mathbb{I}\big(y_{i,t-1} = k\big) + \sum_{l=2}^{4} \beta_{t \times yprev2,l} \cdot t \cdot \mathbb{I}\big(y_{i,t-2} = l\big) \\
&\quad + \sum_{j=2}^{3} \beta_{ecog,j} \cdot \text{ecog}_{ij} + \sum_{k=2}^{5} \beta_{diag,k} \cdot \text{diag}_{ik} + f_1(\text{albumin}_i) + f_2(\text{crp}_i)
\end{aligned}
$$

where :

$$
\begin{aligned}
f(t) &= \sum_{p=1}^{3} \beta_{time,p} \cdot B_p(t) \\
f_1(\text{albumin}_i) &= \sum_{m=1}^{2} \beta_{alb,m} \cdot B_{m,\ \text{alb}}(\text{albumin}_i) \\
f_2(\text{crp}_i) &= \sum_{n=1}^{2} \beta_{crp,n} \cdot B_{n,\ \text{crp}}(\text{crp}_i)
\end{aligned}
$$

$$
\begin{aligned}
\pi_1, ..., \pi_K &\sim \text{Dirichlet } (0.392) \text{ , which induces priors on } \ \alpha_j \\
\beta_{\text{tx}} &\sim \mathcal{N}(0, 0.541) \\
\beta_{time,p \in 1,2,3} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{yprev1},k \in 2..4} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{yprev2},l \in 2..4} &\sim \mathcal{N}(0, 100) \\
\beta_{t \times yprev1,k \in 2..4} &\sim \mathcal{N}(0, 100) \\
\beta_{t \times yprev2,l \in 2..4} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{ecog},j \in 2,3} &\sim \mathcal{N}(0, 100) \\
\beta_{\text{diag},k \in 2..5} &\sim \mathcal{N}(0, 100) \\
\beta_{alb,m \in 1,2} &\sim \mathcal{N}(0, 100) \\
\beta_{crp,n \in 1,2} &\sim \mathcal{N}(0, 100) \\
\gamma_{0i} &\sim \mathcal{N}\big(0, \sigma_\gamma\big) \\
\sigma_\gamma &\sim \text{Half-t } (df = 4, \mu = 0, \sigma = 1)
\end{aligned}
$$

Where:

- α_j: Category-specific intercepts (cutpoints).
- η_{it}: Linear predictor.
  - β_{tx}: Main effect of treatment selection (A=1 vs A=0).
  - $f(t)$: A flexible function of study time, modeled as a restricted cubic spline with 4 knots. This generates 3 basis functions ($B_p(t)$)
  - $\beta_{\text{yprev1, k}}$, $\beta_{\text{yprev2, l}}$: Effects of the state at week $t-1$ and $t-2$ (categorical, state 1 reference). The summation stops at 4, as 5 is an absorbing state.
  - $\beta_{t \times yprev1}$, $\beta_{t \times \text{ yprev2}}$: Interaction terms allowing the effect of each previous state to change linearly over time.

  - Baseline covariates: Effects of baseline ECOG, diagnosis, albumin, and C-reactive protein. The continuous covariates albumin and CRP are modeled flexibly using restricted cubic splines with 3 knots (2 basis functions each).
- γ_{it}: A patient-specific random intercept to account for within-patient correlation beyond the Markov dependency.

7.3.5.1 Prior Specifications

- **Intercepts** (α_j): Priors are induced by a Dirichlet(0.392) distribution on the baseline cell probabilities, as recommended by the `rmsb` package for 5 outcome levels.
- Treatment Parameter ($\\beta_{tx}$): We used prior predictive simulation to select a prior for β_{tx} that implies a clinically plausible distribution for our estimand τ_{TAOOH}. Our simulation, based on historical data, suggested that a prior of $\mathcal{N}(0, 0.541)$ for β_{tx} produces a prior distribution for τ_{TAOOH} with a median at 0 weeks (no difference). This prior places 80% of its mass on β_{tx} between $\log(0.5) \approx -0.693$ and $\log(2.0) \approx +0.693$, which our simulations mapped to an asymmetric 80% prior belief interval for τ_{TAOOH} of approximately $[-6.0, +3.0]$.
- **Fixed Effects Coefficients** (β): All fixed-effect regression coefficients are assigned a weakly informative Normal(0, 100) prior (default in `rmsb`).
- **Random Effect Standard Deviation** (σ_γ): The standard deviation of the random intercepts is given a Half-Normal(0, 1) prior (default in `rmsb`).

7.3.5.2 Rationale for selecting the second-order model and interactions

Model comparisons on historical weekly state data showed that a second-order Markov specification meaningfully improved fit (substantial AIC reduction) compared with first-order models, indicating that the state two weeks prior contains additional predictive information. Including interactions between the previous states (yprev, ypprev) and the linear component of time produced further large improvements in fit and captures clinically plausible time-varying effects of recent hospital state. A linearly constrained non-proportional odds for time offered little additional gain while causing convergence problems. For these reasons we adopted a second-order cumulative logit model with yprev and ypprev interactions with the linear week term, retaining a full proportional-odds structure.[16]

7.3.5.3 Rationale for Random Intercepts

Longitudinal data, such as the weekly states, are characterized by within-patient correlation. The second-order Markov model accounts for a portion of this correlation by conditioning on the previous two states. However, residual correlation may remain due to unobserved patient-level heterogeneity or the misspecification of covariate effects (e.g., omission of a predictive covariate or modeling a non-linear effect as linear). Our own (unpublished) simulation studies indicate that such unmodelled correlation can inflate the frequentist Type I error rate, even when using Bayesian methods with common decision thresholds (e.g., P(benefit)>0.95). While frequentist analyses might address this with cluster-robust standard errors, the analogous approach in our Bayesian framework is to include a patient-specific random intercept. Therefore, the TAOOH model includes a random intercept (γ0i) to capture this residual patient-level correlation, aiming to maintain correct Type I error control and provide more robust inference.

7.3.5.4 Implementation

The model will be fitted using the `blrm` function from the `rmsb` package, including the `cluster(id)` term for random effects. MCMC settings as per OS analysis.

```r
library(rmsb)
```

[16]Detailed model fits, AIC comparisons and diagnostic notes can be found here.

```
model_taooh <- blrm(
    formula = y ~ tx + rcs(week, 4) + yprev + ypprev +
                week %ia% yprev + week %ia% ypprev +
                ecog_fstcnt + diagnosis + rcs(albumin, 3) + rcs(c_reactive_protein, 3)
+
                cluster(id),
    data = data_toh,
    iter = 4000, chains = 4, seed = 68818,
)
```

7.3.5.5 Obtaining Estimand (TAOOH)

1. **Fit Model:** Fit the `model_taooh` to each multiply imputed dataset (using the full ITT population).
2. **Calculate Weekly SOPs:** For each posterior draw, calculate the SOPs, $P(Y_{it}^a = j)$, for each week $t = 1, ..., 26$, state $j = 1, ..., 5$, and treatment arm $a \in \{0, 1\}$. This requires a recursive calculation accounting for the second-order dependency and marginalizing over baseline covariates, baseline state pairs (y_{-1}, y_0), and random effects. Deatils can be found in .
3. **Calculate Expected Weeks Alive/Out-of-Hospital per Draw:** For each posterior draw, calculate the expected number of weeks in State 1 for each treatment arm by summing the SOPs for State 1 over the 26 weeks:

$$\mathbb{E}[W^a]_{\text{draw}} = \sum_{t=1}^{26} \text{SOP } (Y_{it}^a = 1)_{\text{draw}}$$

4. **Calculate Difference per Draw:** For each posterior draw, compute the difference $\tau_{TAOOH,\text{ draw}} = \mathbb{E}[W^1]_{\text{draw}} - \mathbb{E}[W^0]_{\text{draw}}$.
5. **Summarize Posterior:** The collection of $\tau_{TAOOH,\text{ draw}}$ values forms the posterior distribution of the primary estimand for TAOOH.

7.3.5.6 Model Diagnostics: Assessment of Correlation Structure

To validate that our model adequately captures the complex longitudinal dependency in the TAOOH data, we will perform graphical checks. We will compare the empirical correlation structure of the observed weekly states against the correlation structure of data simulated from the fitted model. This comparison will be visualized using Spearman correlation heatmaps (Figure 4) and variograms (Figure 5). This diagnostic check is possible for the TAOOH outcome because it is measured as a complete weekly series; a similar check is not feasible for the QoL endpoint due to its sparse and irregular data collection. The code used for the diagnostics can be found here.

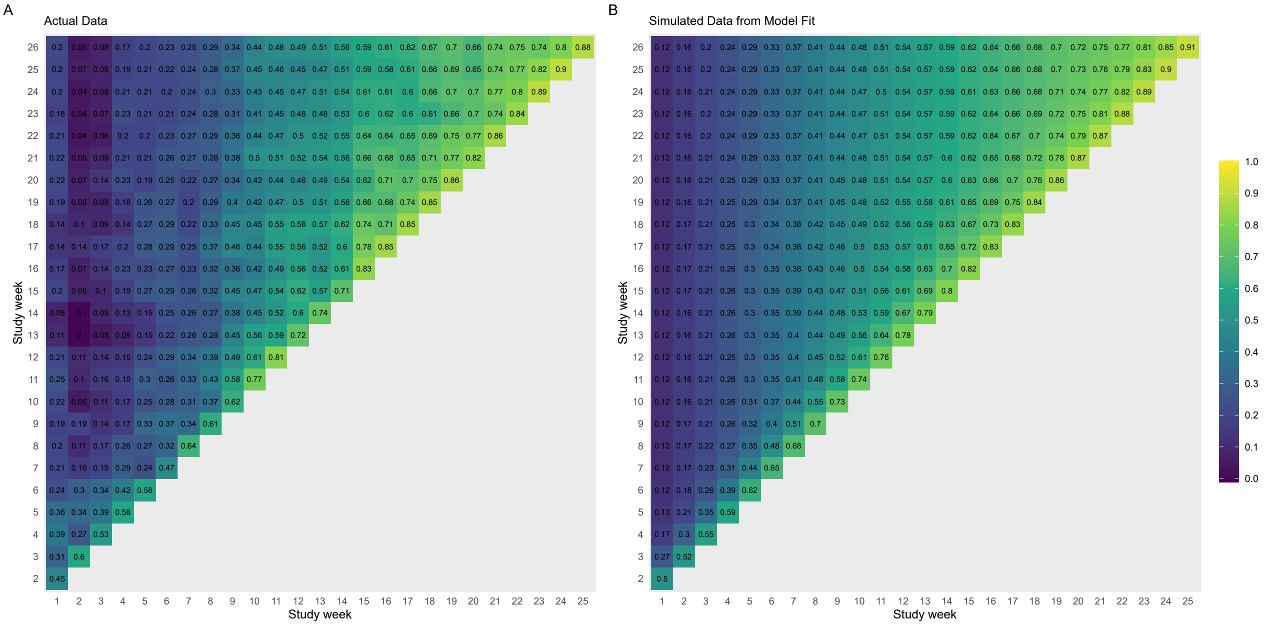


Figure 4: The left plot (A) shows the Spearman correlation between outcomes on each par of study weeks using historical data. The correlation is high for nearby weeks, but quickly decays during the first 9 weeks. The right plot (B) uses simulated data from the fitted model, resampling baseline covariates. We observe a decent match, indicating that our model is able to capture the correlation structure.

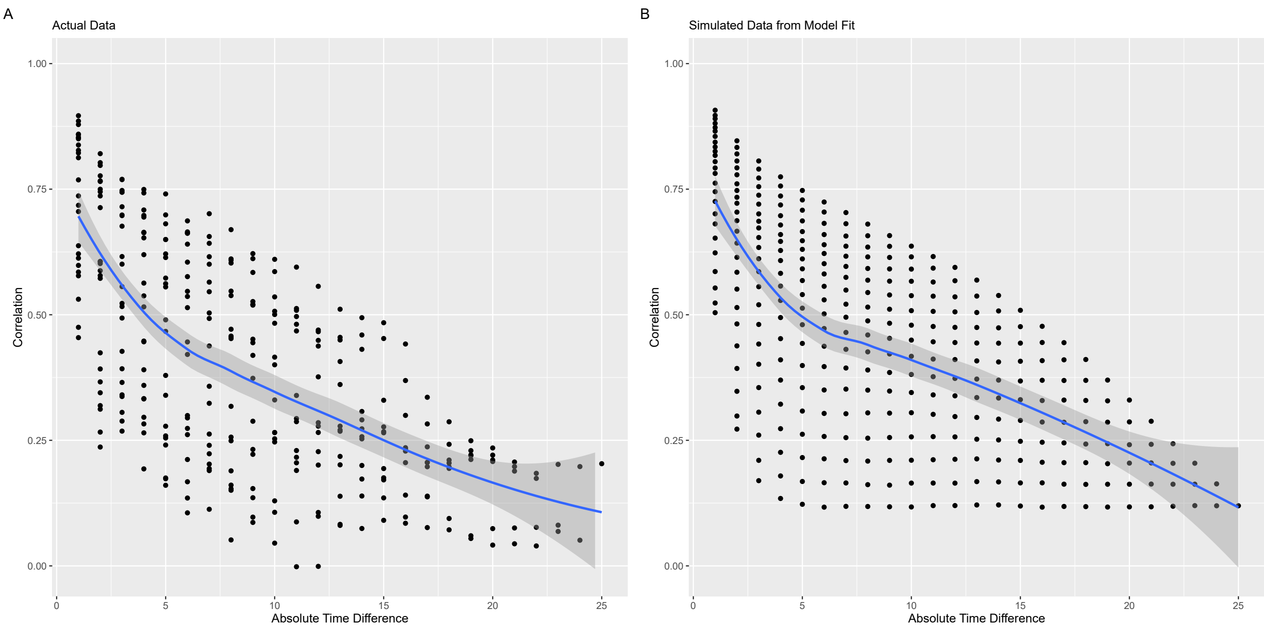


Figure 5: Comparison of variograms, which plot the mean Spearman correlation (y-axis) against the time lag in weeks between measurements (x-axis). Panel (A), using historical data, shows the empirical correlation decaying as the time lag increases. Panel (B) shows the same plot for data simulated from the fitted model. The close alignment between the two plots suggests that the model effectively captures the decay in correlation over time.

7.3.5.7 Frequentist Operating Characteristics (TAOOH)

Using models fit on historical data, we generated datasets with known treatment effects, $OR \in \{0.0, 0.7, 0.8, 0.9\}$. In our historical data, this approximately equaled the following differences in the weeks spent at home and alive over 26 weeks $\tau_{TAOOH} \in \{0, 0.6, 1.2, 1.8\}$. More information on the data generation can be found here.

True τ	OR	Power
~0.6 weeks	0.9	0.238 (0.013)
~1.2 weeks	0.8	0.598 (0.016)
~1.8 weeks	0.7	0.905 (0.009)

Table 2: Bayesian Power when $P(\text{benefit}) > 0.95$ is chosen as a cut-off for benefit under different true odds ratios (Monte Carlo SE in parenthesis).

We emphasize that the $P(\text{benefit}) > 0.95$ threshold is used here only for the purpose of evaluating operating characteristics. It will not be used as a formal, dichotomous decision rule in the final analysis. Instead, the primary reporting will focus on presenting and interpreting the full posterior distribution for the estimand, τ_{TAOOH}. The power figures in Table 2 are intended to provide a conventional benchmark to illustrate which effect sizes the study can reasonably detect given our sample size and modeling approach.

8 Limitations

8.1 QoL Analysis Limitations

8.1.1 Principal Stratum Assumption:

The primary QoL analysis relies on the strong, untestable assumption that selection for invitation has no effect on 6-month survival. If this assumption is violated, the analysis results apply only to the specific subgroup who would survive regardless, but this subgroup is not fully identifiable, and the estimate may be biased for the broader ITT population.

8.1.2 Informative Missingness

A primary limitation arises because the treatment strategy is expected to influence post-randomization clinical decisions, which in turn affect whether the QoL outcome is observed (see Figure 6). These decisions affect both future QoL and the probability that QoL is measured, since data is collected during routine care appointments. For instance, a decision to switch a patient to best supportive care (BSC) may lead to fewer clinic visits, resulting in the cessation of QoL data collection.

As a simple illustrative example, assume ctDNA guided management only improves QoL through an early switch to BSC of an ineffective treatment. Let's also assume that all the patients who are switched to BSC have no more follow-up visits, and their improved QoL is not recorded. Logically, this results in an inability to show the true treatment effect.

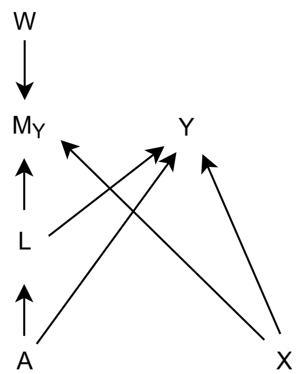


Figure 6: A missingness DAG (m-DAG) depicting assumptions regarding missingness in the quality of life outcome (Y). M_Y is the missingness indicator for Y. W is the vector of unmeasured causes of M_Y. A

represents the selection for invitation to LIQPLAT (treatment indicator), X represents measured baseline covariates, L represents effects of A which affect both follow-up and thus missingness (M_Y) and QoL (Y), such as early switch to best supportive care. $A \to Y$ represent effects of A on Y which do not affect missingness, though we expect these to be rare. Realistically, ctDNA sampling would mostly affect Y through changes in the treatment regimen, which also affects missingness.

8.2 General Limitations

- **Single Center**
- **Intervention Complexity:** The "treatment" is selection for invitation to a complex intervention (ctDNA testing influencing care). Isolating the specific effect of ctDNA itself versus other aspects of trial participation is difficult.

9 Reporting

Results will be reported according to CONSORT guidelines where applicable, adapted for the SAT design. Analyses will follow the ITT principle (based on selection status). Posterior distributions for primary estimands will be visualized (e.g., density plots) and summarized (median, 95% CrI). [Details TBD].

10 References

11 Appendices

11.1 Appendix A: QoL Interval Censoring Investigation

11.1.1 Background

An alternative to the Principal Stratum analysis for QoL is to model the 8-state process (7 QoL states + Death) directly, treating unobserved weeks for living patients as interval-censored (state is in `[1, 7]`). The `rmsb` package supports this via `Ocens()`.

11.1.2 Method

We tested this using a simulated dataset derived from historical data (see `qol-sap.qmd`, Appendix C), where the true state trajectories were known. We created a sparse dataset (15% observations retained post-baseline) and fitted a first-order Markov model using `Ocens(y.a, y.b)` where `y.a=1, y.b=7` for missing weeks of alive patients, and `y.a=y.b=8` at death.

11.1.3 Findings

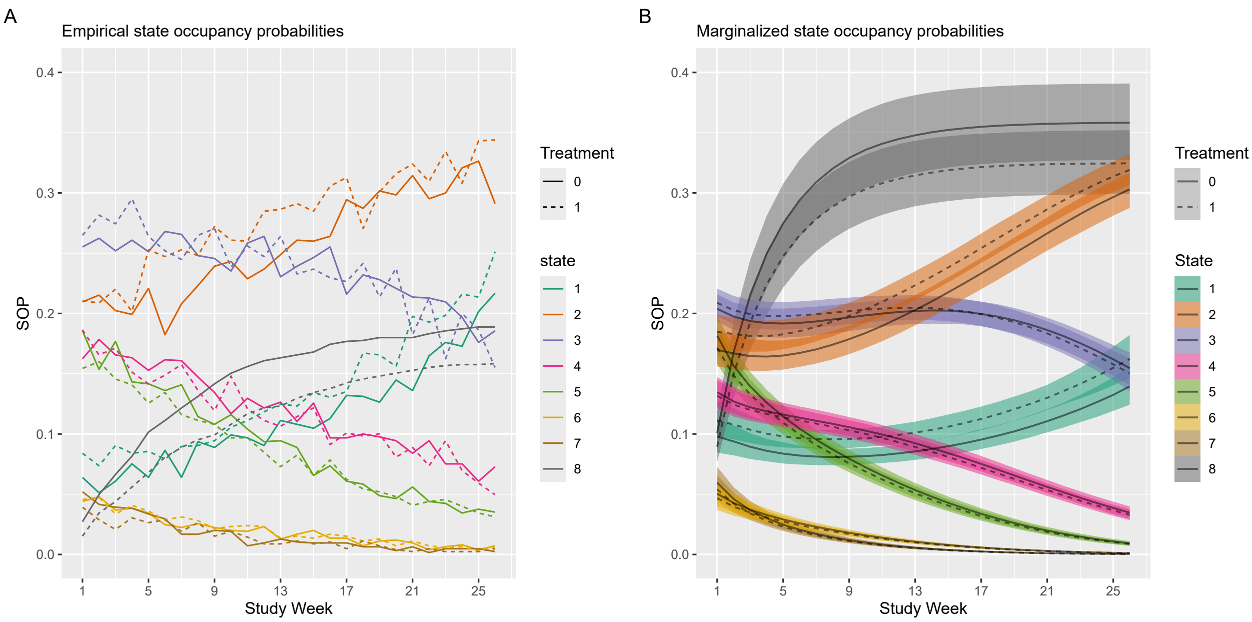


Figure 7: Comparison of empirical SOPs (A) vs. model-derived SOPs using interval censoring (B). The model significantly overestimates the cumulative incidence of death (State 8).

The interval censoring approach led to a substantial overestimation of the cumulative incidence of death (Figure 7). This bias likely arises because the absorbing death state is always observed precisely, while the intermediate living states are heavily censored. The model appears to incorrectly assign probability mass over time towards the known absorbing state. This bias propagates, underestimating time spent in living states.

11.1.4 Conclusion

Due to this potential for significant bias with sparse intermediate data, the interval censoring approach was deemed unsuitable for the primary QoL analysis, leading to the adoption of the Principal Stratum approach with its specific assumptions.

11.2 Appendix B: TAAOH State Occupancy Probabilities

The state occupancy probability (SOP) is a foundational derived quantity from a transition model. For a second-order process, we define the SOP as $P(Y_j = k \mid X, y_{-1}, y_0)$, which is the probability of being in state k at time j, conditional on a set of baseline covariates X and the two baseline states, y_{-1} (state at time $j = -1$) and y_0 (state at time $j = 0$).

In a first-order model, a single transition matrix is sufficient to describe the evolution of the system from time $j-1$ to j (M. D. Rohde, B. French, T. G. Stewart, and F. E. Harrell [14]). In the second-order case, however, the probabilities of transitioning from state k at time $j-1$ to state l at time j are themselves conditional on the state at time $j-2$. This dependency requires a set of transition matrices, one for each possible prior state ($Y_{j-2} = h$), rather than a single matrix. The core challenge is that the state of the system at any given time is not described by a single previous outcome, but by the joint probability distribution of the last two outcomes. Therefore, to compute the SOPs, we must recursively calculate the evolution of this joint probability distribution over time.

Let the conditional transition probability at time j be denoted as $T_j(l \mid h, k) = P\left(Y_j = l \mid Y_{j-2} = h, Y_{j-1} = k, X\right)$. These probabilities, which represent the chance of moving to state l given the history of being in state h then state k, are calculated directly from the fitted ordinal transition model for a specific covariate profile X.

The unconditionalization process proceeds as follows:

1. **Initialization**: We begin with the known baseline states y_{-1} and y_0. The joint probability of being in states h and k at times $j = -1$ and $j = 0$ respectively, which we denote as $J_0(h, k) = P(Y_{-1} = h, Y_0 = k)$, is initialized as a matrix of zeros with a 1 at the position $(h, k) = (y_{-1}, y_0)$. The initial SOP at time $j = 0$ is therefore known with certainty: $S_0(k) = 1$ for the state $k = y_0$ and 0 otherwise.
2. **Recursive Calculation of Joint Probabilities**: For each subsequent time point $j = 1, 2, ..., d$, we calculate the new joint probability distribution $J_j(k, l) = P\left(Y_{j-1} = k, Y_j = l\right)$ by marginalizing over the state at time $j-2$, denoted by h. Using the law of total probability, the recursion is defined as:

   $$J_j(k, l) = \sum_{h=1}^{K} T_j(l \mid h, k) \times J_{j-1}(h, k)$$

   In this step, we use the joint probabilities for times $j-2$ and $j-1$, J_{j-1}, and the conditional transition probabilities for the current time step, T_j, to compute the joint probabilities for the current and previous time steps, J_j.
3. **Calculation of State Occupancy Probabilities**: Once the joint probability matrix J_j for time j is obtained, the SOP for being in state l at time j, denoted $S_j(l)$, is calculated by marginalizing out the state at time $j-1$ (denoted by k):

   $$S_j(l) = P\left(Y_j = l\right) = \sum_{k=1}^{K} J_j(k, l)$$

This $S_j(l)$ is the marginal probability at time j and is the state occupancy probability $P\left(Y_j = l \mid X, y_{-1}, y_0\right)$ for the given baseline conditions. This three-step process is repeated for all time points up to the desired horizon, yielding the full set of SOPs for a subject with a given covariate profile and specific baseline history. The code that implements these calculations can be found here.

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